

Cost Analysis

2026-07-20

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1 Assumptions

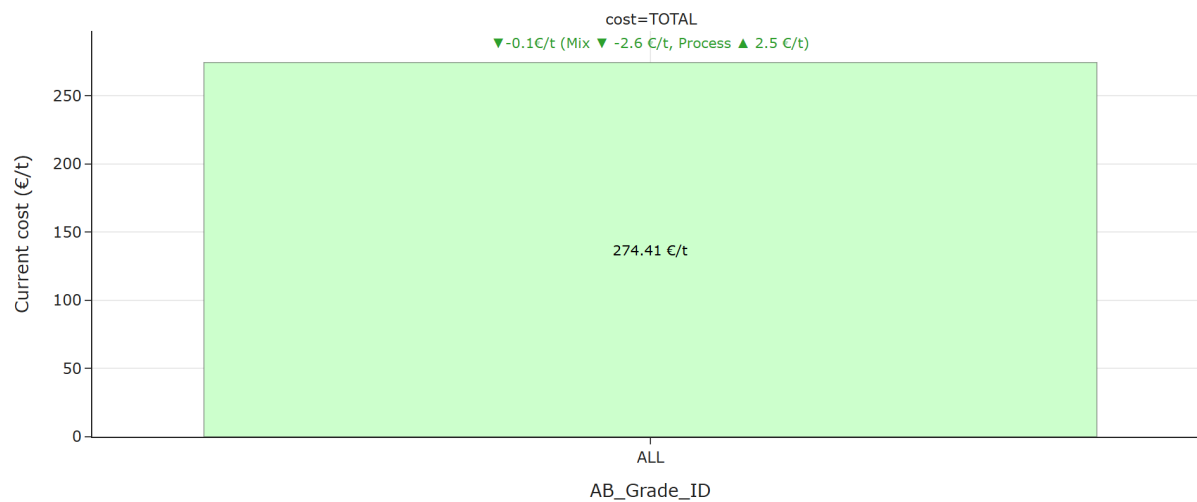
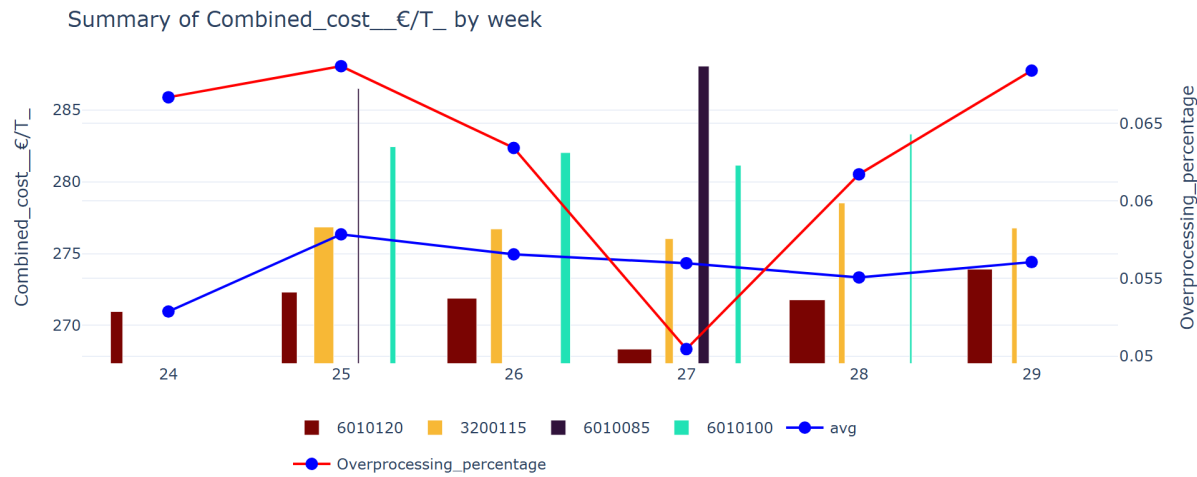
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** June 14, 2026 – July 20, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.
 - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
 - **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

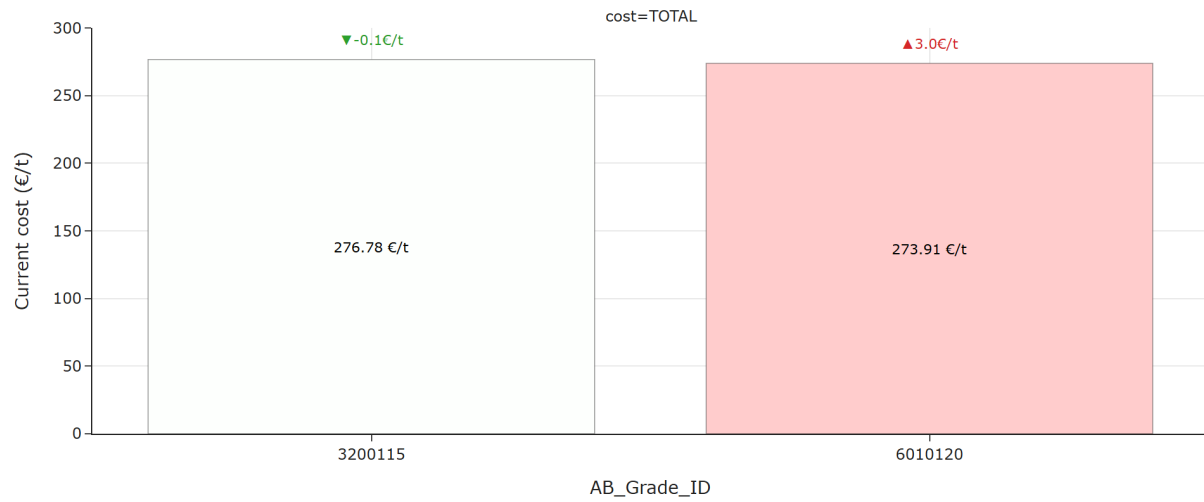
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw a decrease of 0.08 €/t, resulting in 274.41 €/t.

Breaking down the overall cost change, -2.57 €/t was driven by a cheaper mix of grades, and +2.49 €/t was due to a deterioration in the process.



For grade 3200115, TOTAL cost saw a decrease of 0.13 €/t, resulting in 276.78 €/t.

For grade 6010120, TOTAL cost saw an increase of 3.04 €/t, resulting in 273.91 €/t.

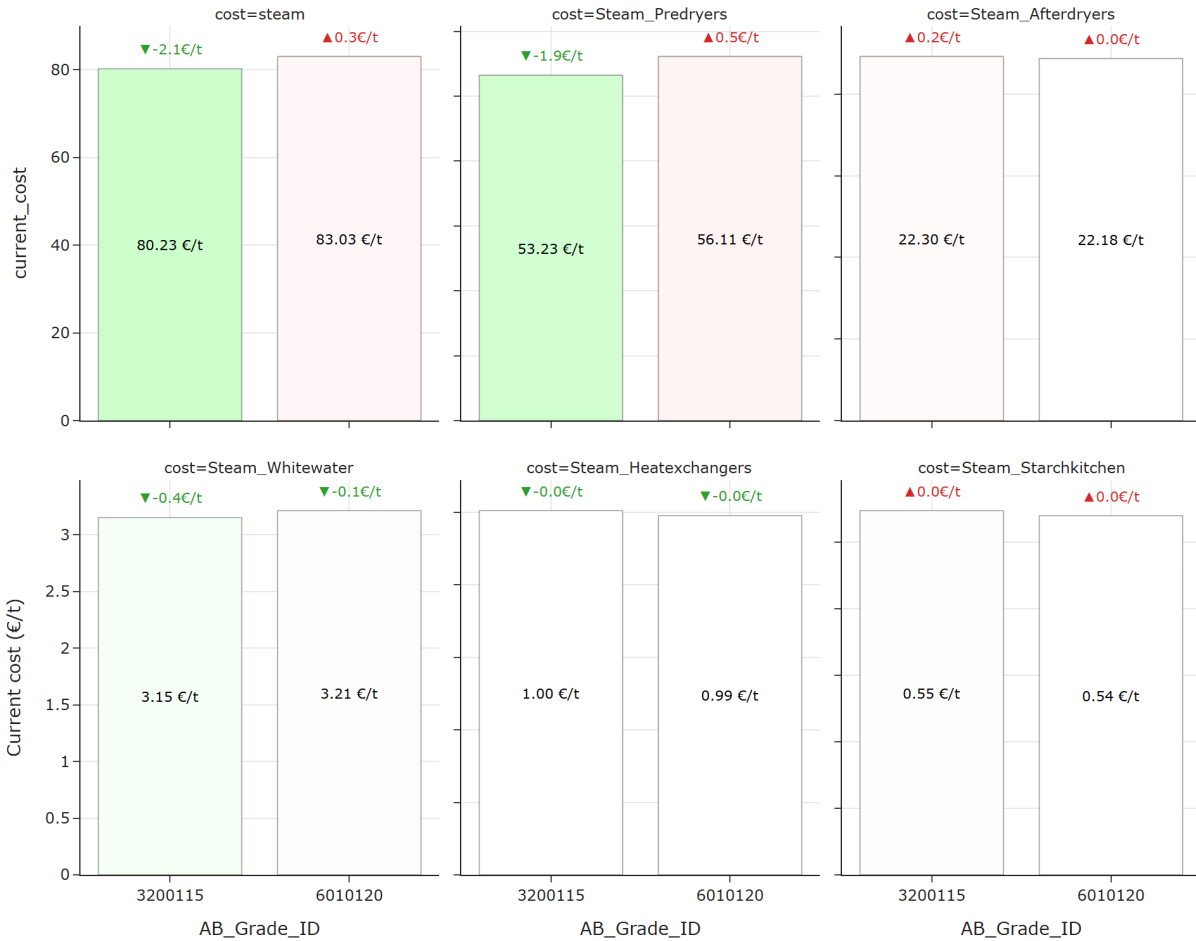
Looking across individual grades, the strongest cost increase was observed for grade 6010120 (+3.04 €/t) and the strongest cost decrease was observed for grade 3200115 (-0.13 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Electricity_€/T** increased on average (+1.01 €/t).
- **Steam_€/T** decreased on average (-0.89 €/t).
- **Starch_€/T** increased on average (+0.83 €/t).
- **Fibre_cost_€/T** increased on average (+0.34 €/t).
- **Chemicals_€/T** increased on average (+0.16 €/t).

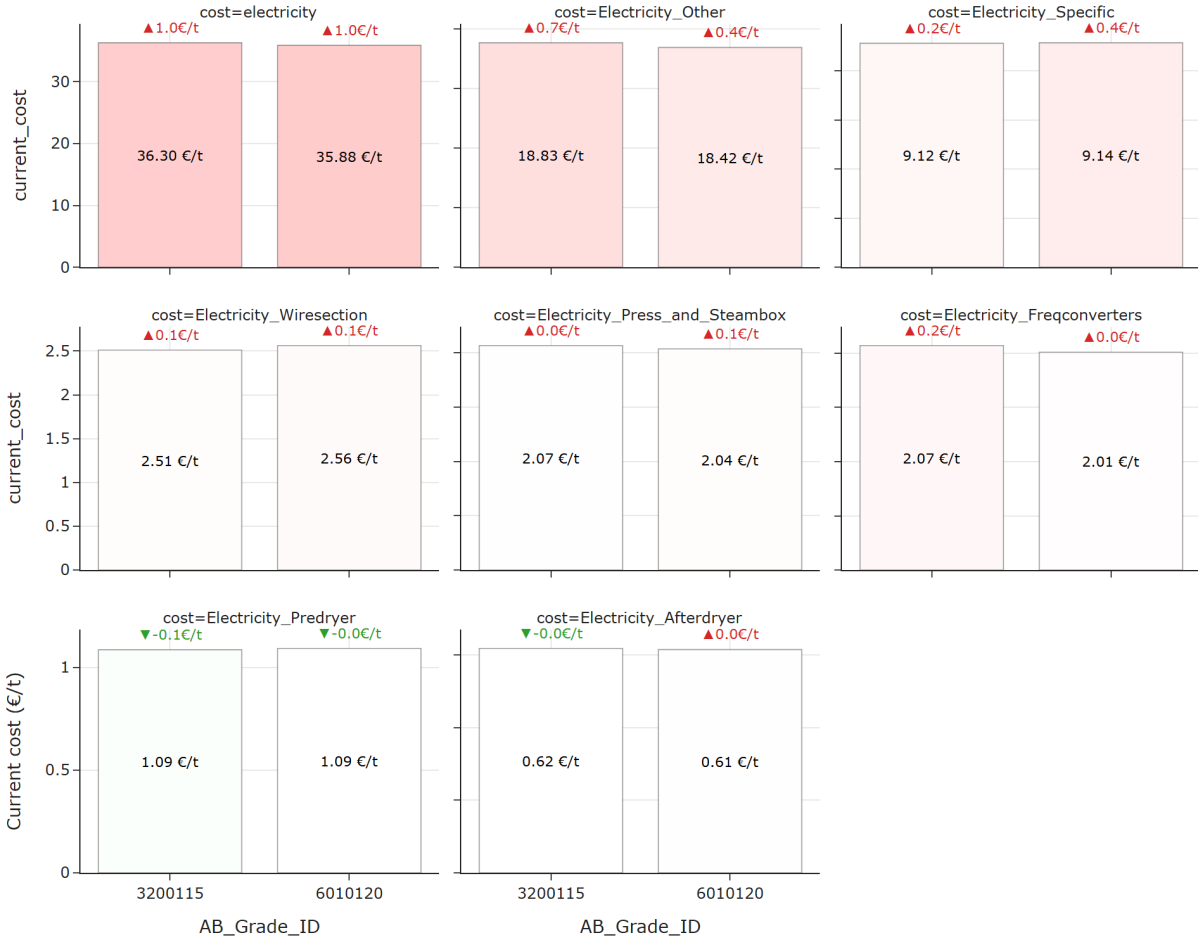
A more detailed breakdown by component shows that the largest average increase comes from **Electricity_€/T** (+1.01 €/t on average), with the highest increase in grade 6010120 (+1.02 €/t) and the largest average decrease comes from **Steam_€/T** (-0.89 €/t on average), with the biggest drop in grade 3200115 (-2.11 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Predryers** decreased on average (-0.72 €/t).
- **Steam_Whitewater** decreased on average (-0.27 €/t).
- **Steam_Afterdryers** increased on average (+0.10 €/t).
- **Steam_Starchkitchen** increased on average (+0.02 €/t).
- **Steam_Heatexchangers** decreased on average (-0.02 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Afterdryers** (+0.10 €/t on average), with the highest increase in grade 3200115 (+0.20 €/t) and the largest average decrease comes from **Steam_Predryers** (-0.72 €/t on average), with the biggest drop in grade 3200115 (-1.91 €/t). These components stand out and may require further investigation.



Average change by electricity:

- **Electricity_Other** increased on average (+0.55 €/t).
- **Electricity_Specific** increased on average (+0.28 €/t).
- **Electricity_Freqconverters** increased on average (+0.10 €/t).
- **Electricity_Wiresection** increased on average (+0.08 €/t).
- **Electricity_Predryer** decreased on average (-0.04 €/t).
- **Electricity_Press_and_Steambox** increased on average (+0.04 €/t).
- **Electricity_Afterdryer** increased on average (+0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Other** (+0.55 €/t on average), with the highest increase in grade 3200115 (+0.67 €/t) and the largest average decrease comes from **Electricity_Predryer** (-0.04 €/t on average),

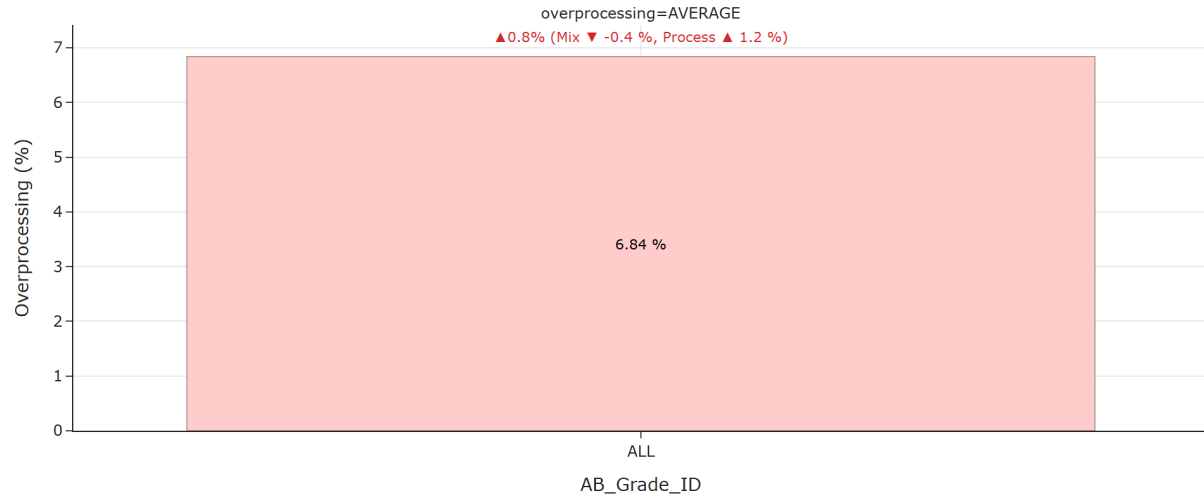
with the biggest drop in grade 3200115 (-0.07 €/t). These components stand out and may require further investigation.



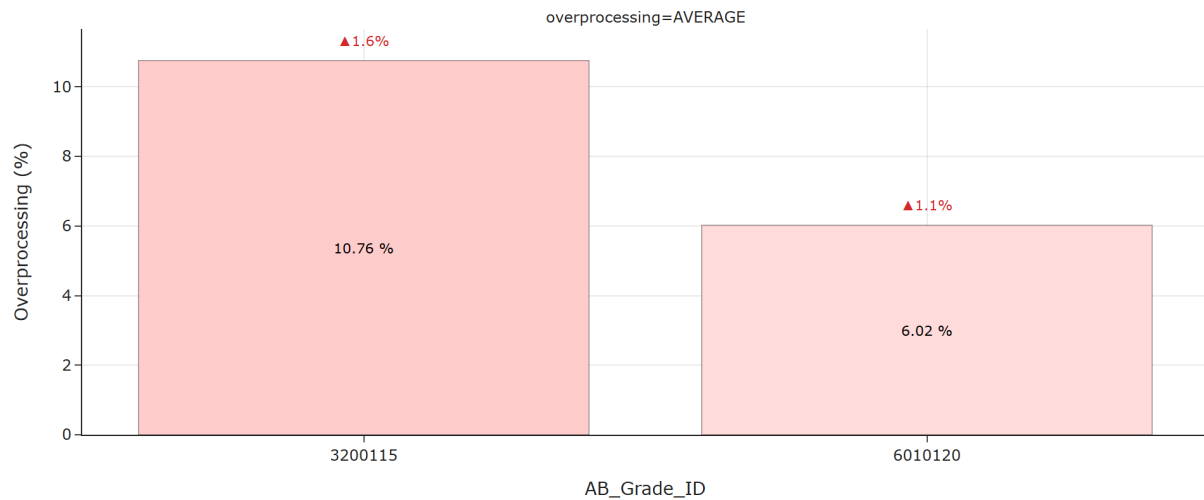
Average change by chemicals:

- **Sizing_Agent_€/T** increased on average (+0.11 €/t).
- **Fixative_2_€/T** increased on average (+0.09 €/t).
- **Bentonite_2_€/T** decreased on average (-0.09 €/t).
- **Retention_Aid_€/T** increased on average (+0.08 €/t).
- **Natriumhydroxide_€/T** decreased on average (-0.05 €/t).
- **Bentonite_1_€/T** increased on average (+0.03 €/t).
- **Defoamer_€/T** decreased on average (-0.02 €/t).
- **Deaerator_€/T** increased on average (+0.02 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Sizing_Agent_€/T** (+0.11 €/t on average), with the highest increase in grade 3200115 (+0.17 €/t) and the largest average decrease comes from **Bentonite_2_€/T** (-0.09 €/t on average), with the biggest drop in grade 3200115 (-0.09 €/t). These components stand out and may require further investigation.



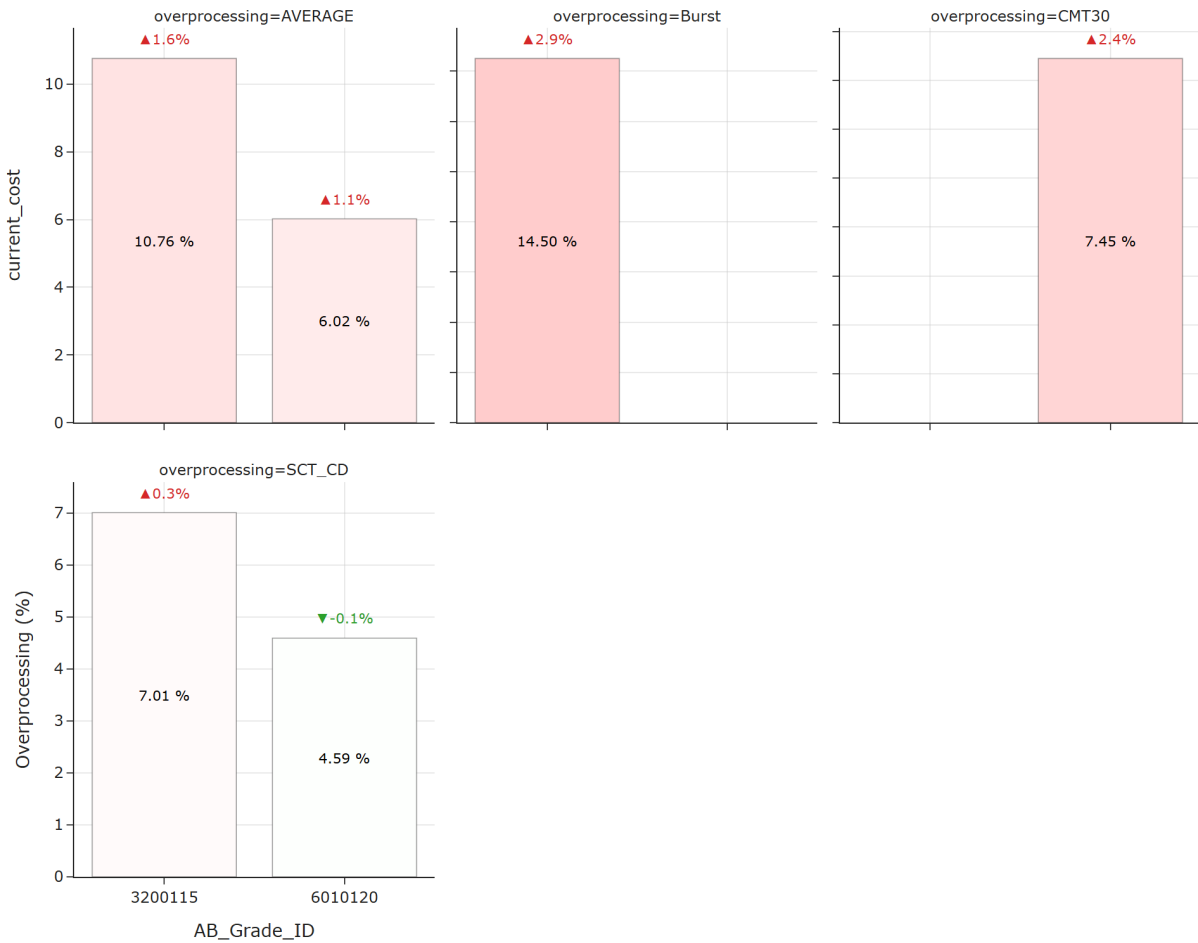
For all considered grades, TOTAL overprocessing saw an increase of 0.77 %, resulting in 6.84 %.



For grade 3200115, TOTAL overprocessing saw an increase of 1.60 %, resulting in 10.76 %.

For grade 6010120, TOTAL overprocessing saw an increase of 1.12 %, resulting in 6.02 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 3200115 (+1.60 %). These movements stand out and may require further investigation.



Average change by component:

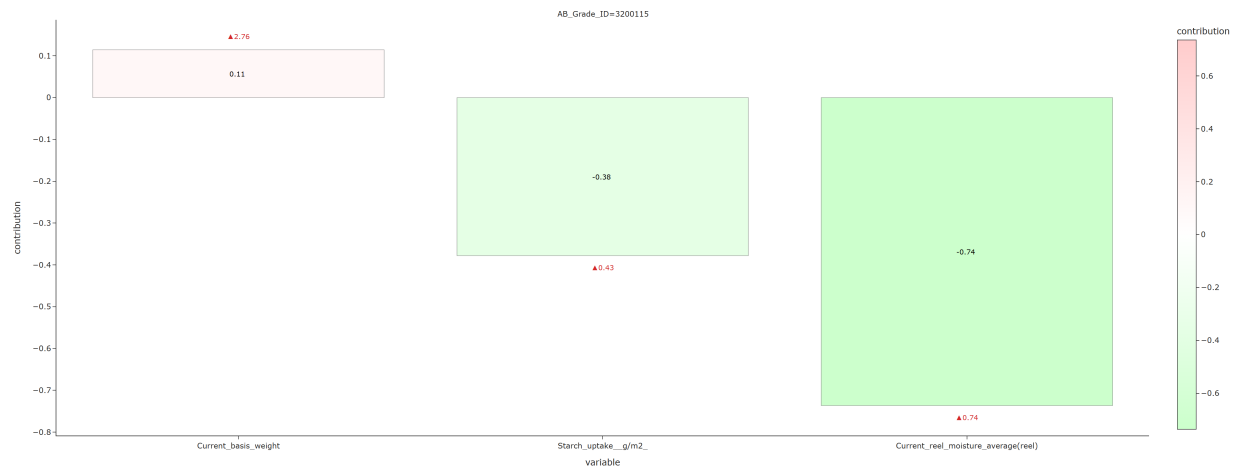
- **Burst** increased on average (+2.92 %).
- **CMT30** increased on average (+2.37 %).
- **SCT_CD** increased on average (+0.08 %).

A more detailed breakdown by component shows that the largest average increase comes from **Burst** (+2.92 % on average), with the highest increase in grade 3200115 (+2.92 %). These components stand out and may require further investigation.

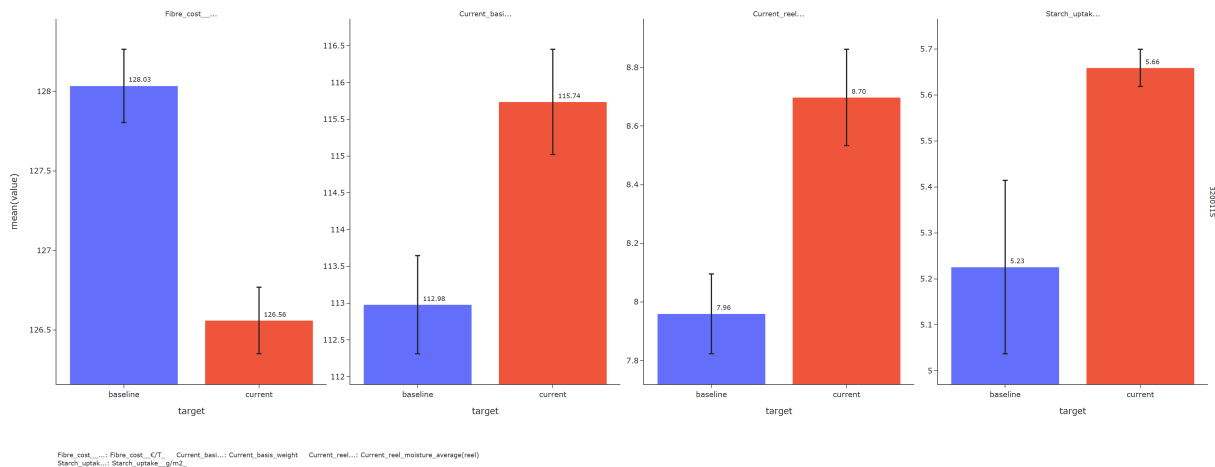
3 Breakdown by grades

3.1 3200115

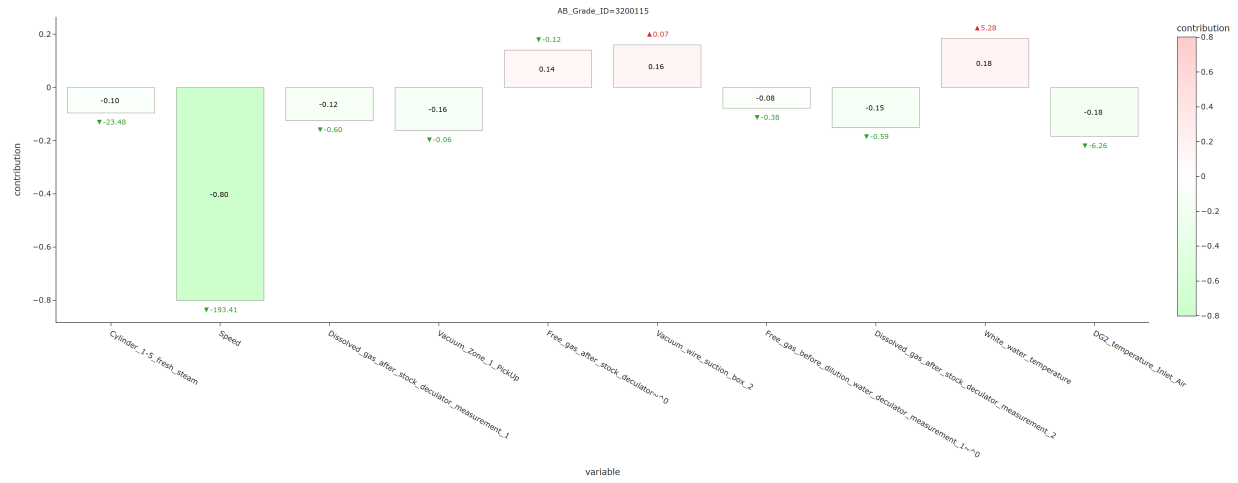
3.1.1 Fiber cost



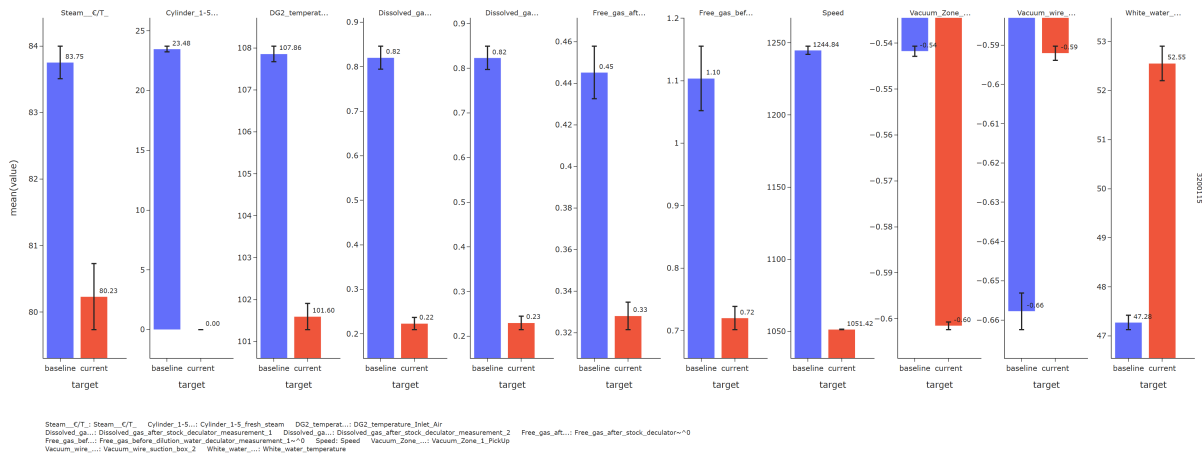
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Current reel moisture average(reel) increased (+0.74)



3.1.2 Steam cost



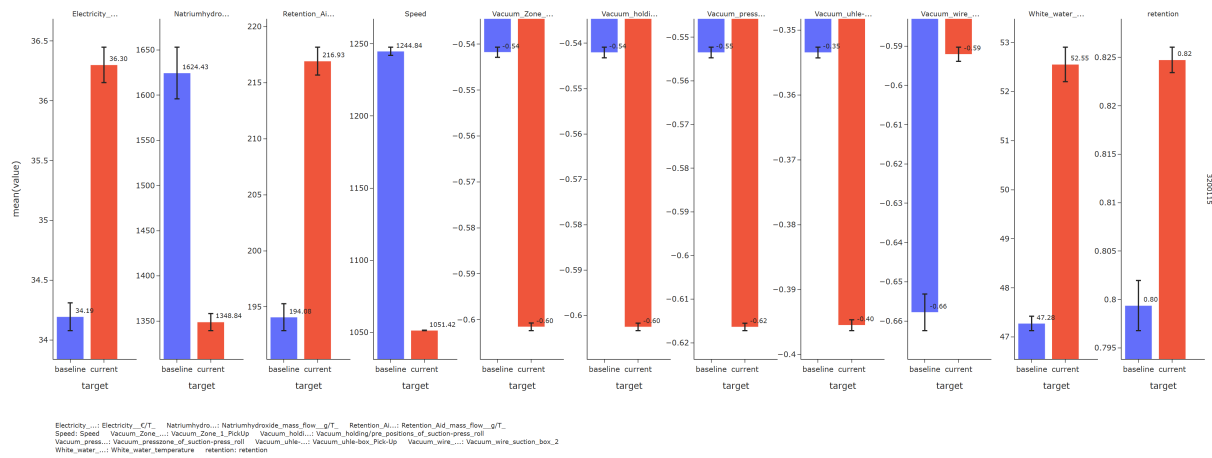
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Speed decreased (-193.41) - DG2 temperature Inlet Air decreased (-6.26) - Vacuum Zone 1 Pickup decreased (-0.06) - Dissolved gas after stock deculator measurement 2 decreased (-0.59) - Dissolved gas after stock deculator measurement 1 decreased (-0.60) - Cylinder 1-5 fresh steam decreased (-23.48) **Cost-increasing drivers:** - White water temperature increased (+5.28) - Vacuum wire suction box 2 increased (+0.07) - Free gas after stock decuator~0 decreased (-0.12)



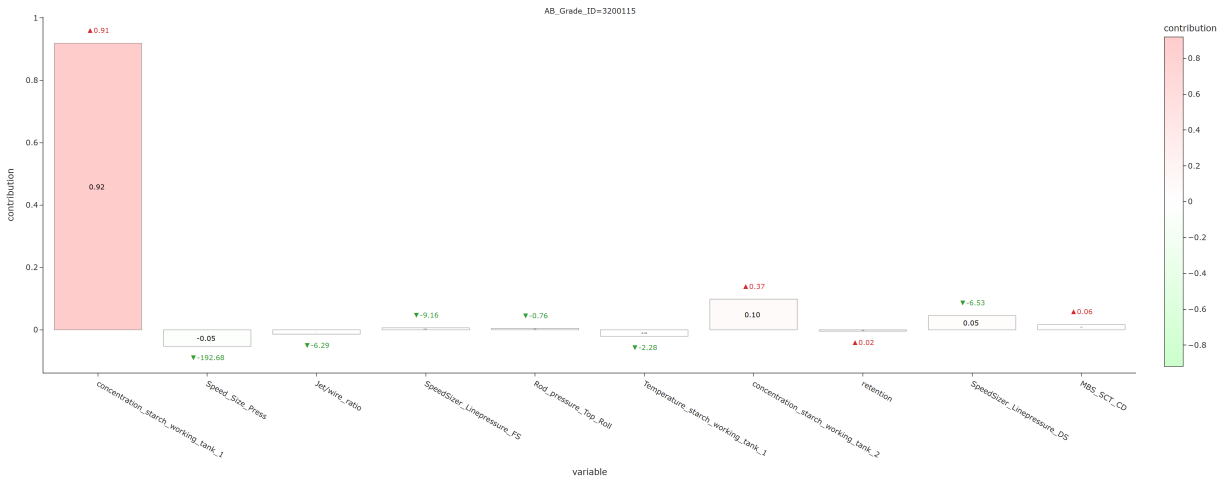
3.1.3 Electricity cost



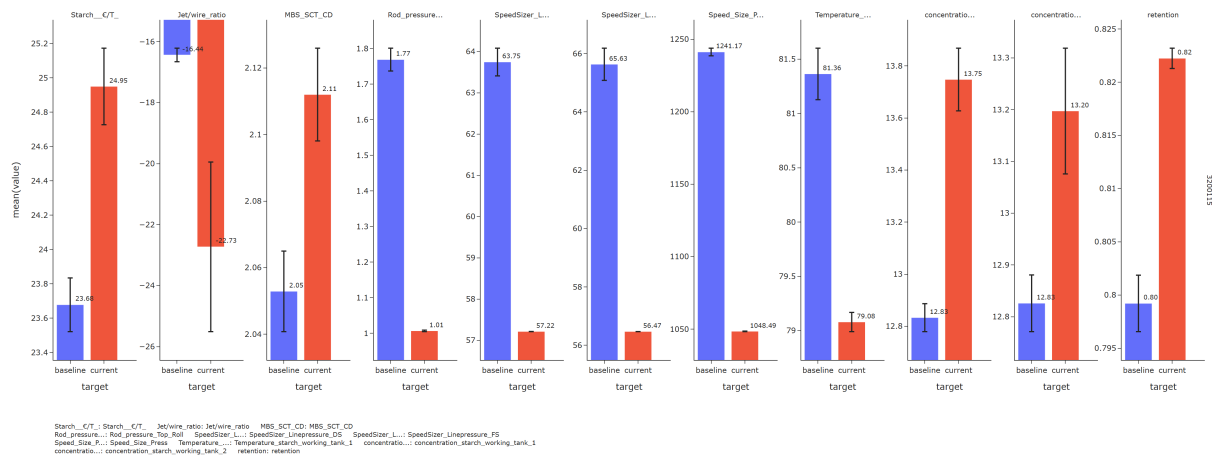
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - White water temperature increased (+5.28) **Cost-increasing drivers:** - Vacuum presszone of suction-press roll decreased (-0.06) - Vacuum holding/pre positions of suction-press roll decreased (-0.06)



3.1.4 Starch cost

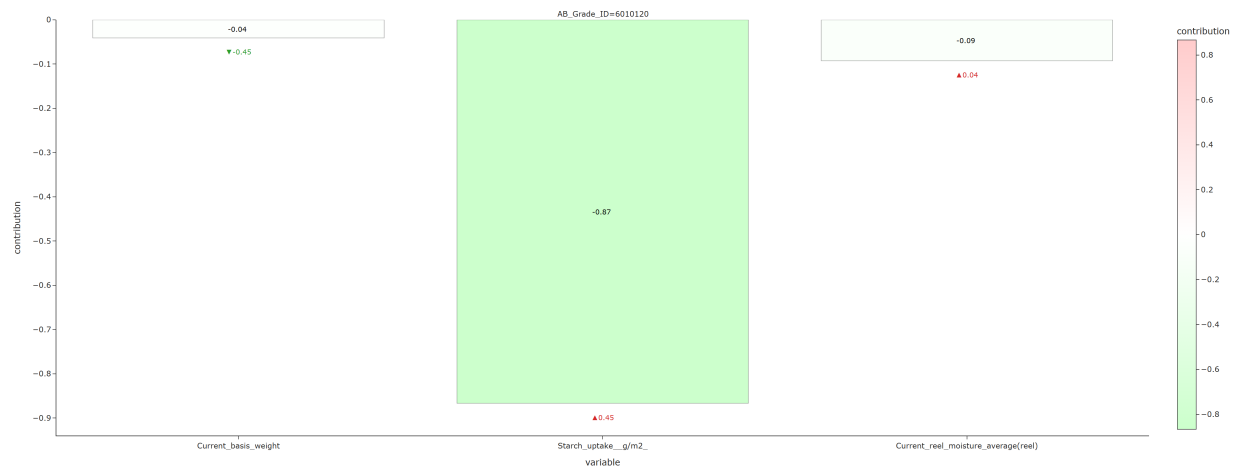


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Speed Size Press decreased (-192.68) **Cost-increasing drivers:** - Concentration starch working tank 1 increased (+0.91) - Concentration starch working tank 2 increased (+0.37)

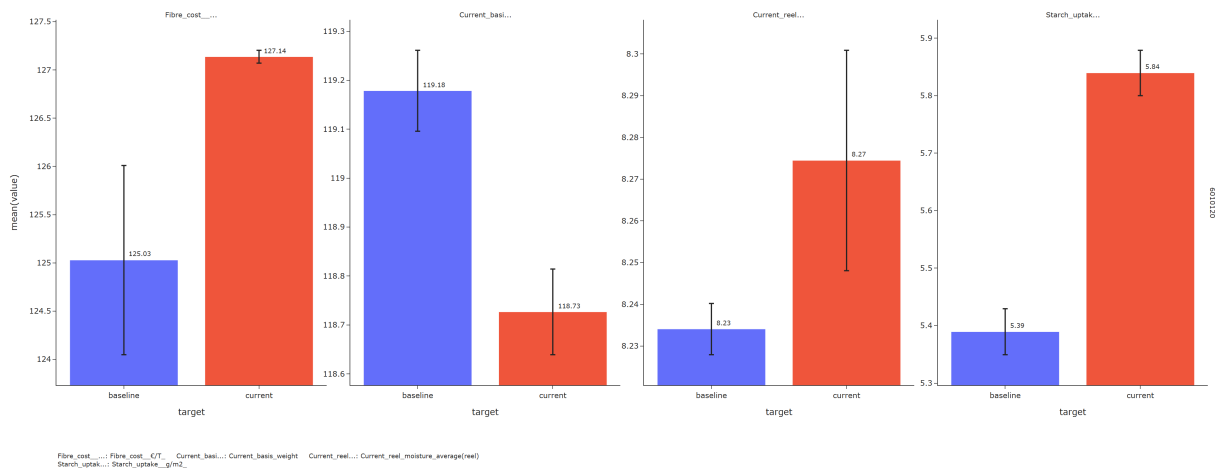


3.2 6010120

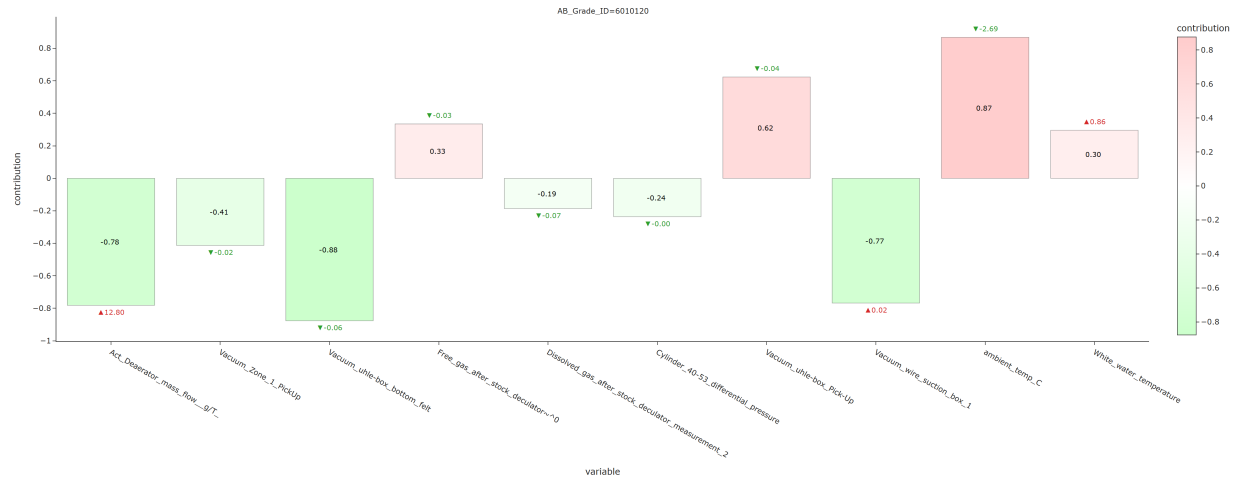
3.2.1 Fiber cost



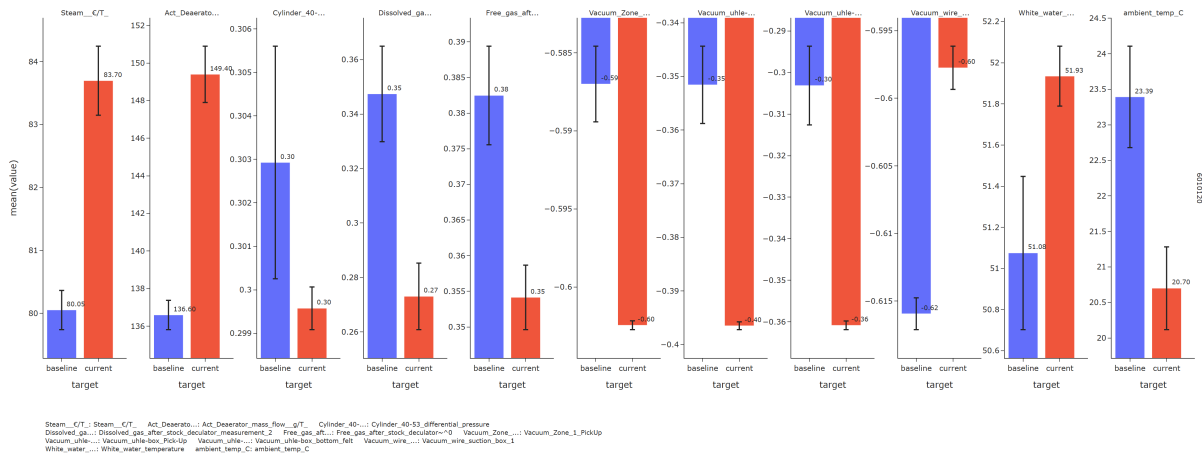
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Starch uptake g/m2 increased (+0.45)



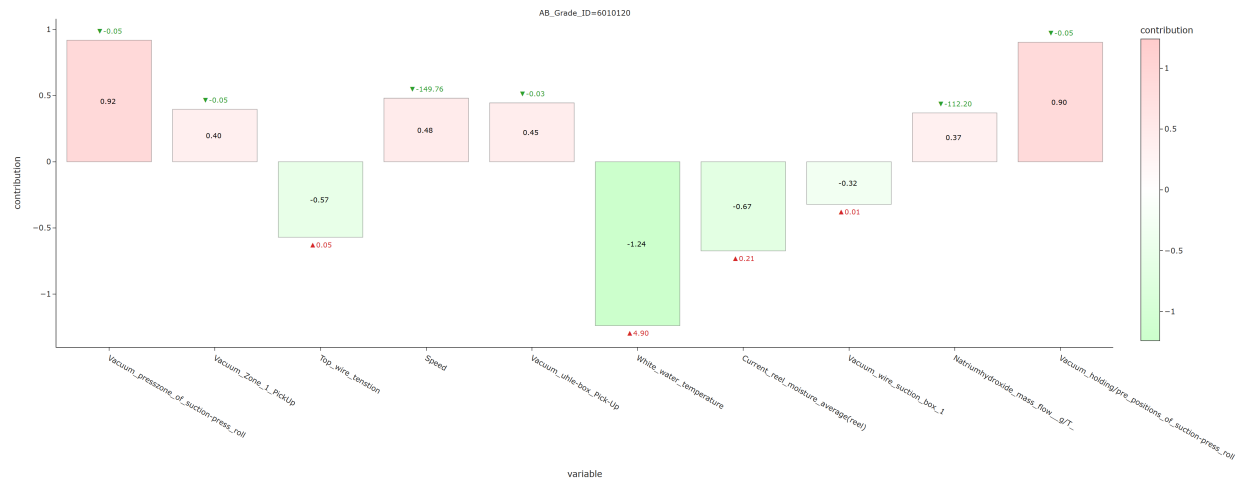
3.2.2 Steam cost



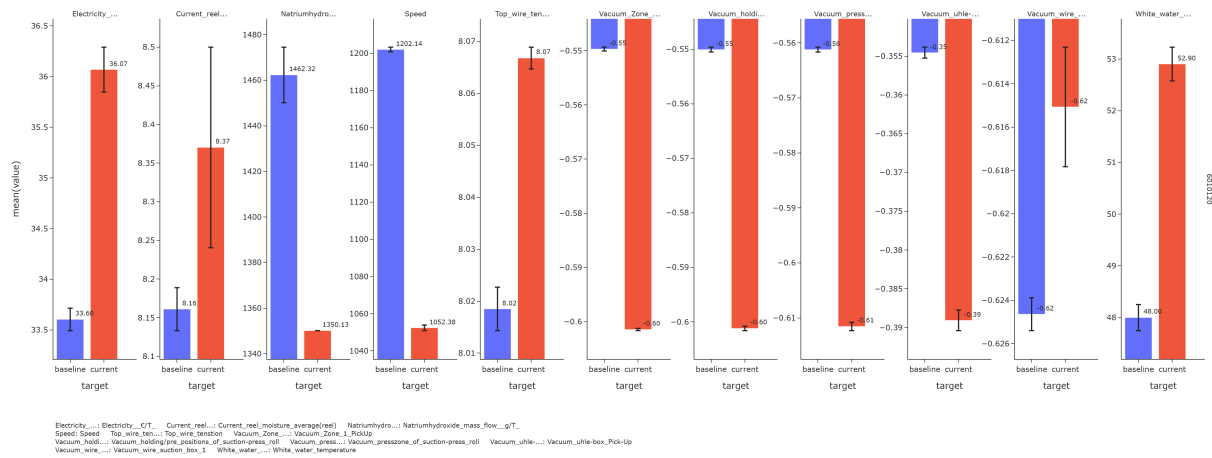
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Vacuum uhle-box bottom felt decreased (-0.06) - Act Deaerator mass flow g/T increased (+12.80) - Vacuum wire suction box 1 increased (+0.02) - Vacuum Zone 1 PickUp decreased (-0.02) **Cost-increasing drivers:** - Ambient temp C decreased (-2.69) - Vacuum uhle-box Pick-Up decreased (-0.04)



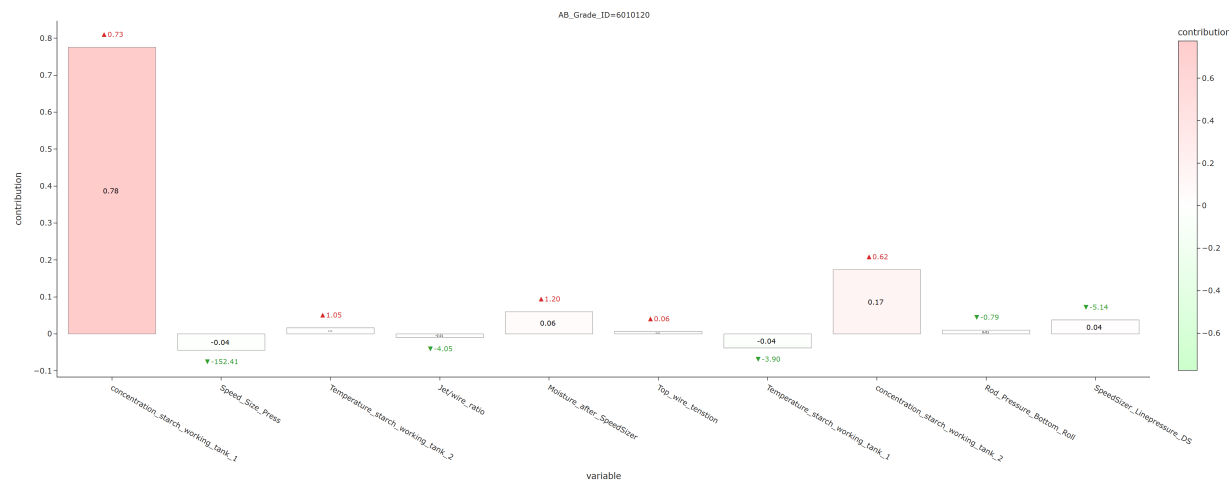
3.2.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - White water temperature increased (+4.90) **Cost-increasing drivers:** - Vacuum presszone of suction-press roll decreased (-0.05) - Vacuum holding/pre positions of suction-press roll decreased (-0.05)



3.2.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Concentration starch working tank 1 increased (+0.73) - Concentration starch working tank 2 increased (+0.62) - Moisture after SpeedSizer increased (+1.20)

